



PREMIUM

Full-Length Mock Test

Mock 03

Instrumentation Engineering

GATE IN

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20%

Moderate 33%

Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	35	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		60		100	

Marking Scheme

Correct MCQ: +1 mark (or +2 marks as specified per question)

Wrong MCQ: -0.33 marks

Correct MSQ: +2 marks

Wrong MSQ (partial or no match): -0.33 marks

Correct NAT: +1 mark (or +2 marks as specified per question)

No answer / Unattempted: 0 marks

NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Sensors & Transducers — Resistive/Capacitive]

LVDT measures:

- A) Displacement
 - B) Velocity
 - C) Acceleration
 - D) Force
-

Q.2 [1 Mark] [EASY] [Measurement Systems — Bridge Circuits]

RTD uses:

- A) Semiconductor
 - B) Pure metal
 - C) Thermocouple
 - D) Thermistor
-

Q.3 [1 Mark] [EASY] [Control Systems — Compensators]

Strain gauge factor $GF =$:

- A) $\frac{\Delta L}{L}$
 - B) $\frac{\Delta R}{R}$
 - C) $\frac{\Delta R}{R} \times \frac{L}{\Delta L}$
 - D) $\frac{\Delta L}{\Delta R}$
-

Q.4 [1 Mark] [EASY] [Digital Electronics — ADC/DAC]

CRO displays:

- A) Current vs Time
 - B) Voltage vs Time
 - C) Power vs Time
 - D) Frequency vs Time
-

Q.5 [1 Mark] [EASY] [Analog Electronics — Oscillators]

LVDT measures:

- A) Displacement
 - B) Velocity
 - C) Acceleration
 - D) Force
-

Q.6 [1 Mark] [EASY] [Sensors & Transducers — Resistive/Capacitive]

RTD uses:

- A) Semiconductor
 - B) Pure metal
 - C) Thermocouple
 - D) Thermistor
-

Q.7 [1 Mark] [EASY] [Measurement Systems — CRO]

Strain gauge factor $GF =$:

A) 9E"ö" ò "L/L

B) 9DÂôÂ ò "R/R

C) 9E"ö" r" L/L

D) 9E"Ó"L

Q.8 [1 Mark] [EASY] [Control Systems — Root Locus]

CRO displays:

- A) Current vs Time
 - B) Voltage vs Time
 - C) Power vs Time
 - D) Frequency vs Time
-

Q.9 [1 Mark] [EASY] [Digital Electronics — Logic Gates]

LVDT measures:

- A) Displacement
 - B) Velocity
 - C) Acceleration
 - D) Force
-

Q.10 [1 Mark] [EASY] [Analog Electronics — Op-Amps]

RTD uses:

- A) Semiconductor
 - B) Pure metal
 - C) Thermocouple
 - D) Thermistor
-

Q.11 [1 Mark] [EASY] [Sensors & Transducers — Thermoelectric]

Strain gauge factor $GF =$:

- A) $\frac{\Delta L}{L} \div \frac{\Delta R}{R}$
 - B) $\frac{\Delta R}{R} \div \frac{\Delta L}{L}$
 - C) $\frac{\Delta R}{R} \times \frac{\Delta L}{L}$
 - D) $\frac{\Delta L}{L} \times \frac{\Delta R}{R}$
-

Q.12 [1 Mark] [EASY] [Measurement Systems — CRO]

CRO displays:

- A) Current vs Time
 - B) Voltage vs Time
 - C) Power vs Time
 - D) Frequency vs Time
-

Q.13 [2 Marks] [MODERATE] [Control Systems — PID Control]

LVDT measures:

- A) Displacement
 - B) Velocity
 - C) Acceleration
 - D) Force
-

Q.14 [2 Marks] [MODERATE] [Digital Electronics — Sequential Circuits]

RTD uses:

- A) Semiconductor

B) Pure metal

C) Thermocouple

D) Thermistor

Q.15 [2 Marks] [MODERATE] [Analog Electronics — Op-Amps]

Strain gauge factor GF =:

- A) $9E''\delta'' \rightarrow ''L/L$
- B) $9D\hat{A}\hat{A} \rightarrow ''R/R$
- C) $9E''\delta'' \rightarrow ''L/L$
- D) $9E''\delta''L$

Q.16 [2 Marks] [MODERATE] [Sensors & Transducers — Optical Sensors]

CRO displays:

- A) Current vs Time
- B) Voltage vs Time
- C) Power vs Time
- D) Frequency vs Time

Q.17 [2 Marks] [MODERATE] [Measurement Systems — Errors]

LVDT measures:

- A) Displacement
- B) Velocity
- C) Acceleration
- D) Force

Q.18 [2 Marks] [MODERATE] [Control Systems — PID Control]

RTD uses:

- A) Semiconductor
- B) Pure metal
- C) Thermocouple
- D) Thermistor

Q.19 [2 Marks] [MODERATE] [Digital Electronics — Sequential Circuits]

Strain gauge factor GF =:

- A) $9E''\delta'' \rightarrow ''L/L$
- B) $9D\hat{A}\hat{A} \rightarrow ''R/R$
- C) $9E''\delta'' \rightarrow ''L/L$
- D) $9E''\delta''L$

Q.20 [2 Marks] [MODERATE] [Analog Electronics — Diodes]

CRO displays:

- A) Current vs Time
- B) Voltage vs Time
- C) Power vs Time
- D) Frequency vs Time

Q.21 [2 Marks] [MODERATE] [Sensors & Transducers — Thermoelectric]

LVDT measures:

- A) Displacement

B) Velocity

C) Acceleration

D) Force

Q.22 [2 Marks] [MODERATE] [Measurement Systems — Signal Conditioning]

RTD uses:

- A) Semiconductor
 - B) Pure metal
 - C) Thermocouple
 - D) Thermistor
-

Q.23 [2 Marks] [MODERATE] [Control Systems — PID Control]

Strain gauge factor GF =:

- A) $\Delta R/R \div \Delta L/L$
 - B) $\Delta R/R \div \Delta L/L$
 - C) $\Delta R/R \times \Delta L/L$
 - D) $\Delta R/R \times \Delta L$
-

Q.24 [2 Marks] [MODERATE] [Digital Electronics — Logic Gates]

CRO displays:

- A) Current vs Time
 - B) Voltage vs Time
 - C) Power vs Time
 - D) Frequency vs Time
-

Q.25 [2 Marks] [MODERATE] [Analog Electronics — Amplifiers]

LVDT measures:

- A) Displacement
 - B) Velocity
 - C) Acceleration
 - D) Force
-

Q.26 [2 Marks] [MODERATE] [Sensors & Transducers — Thermoelectric]

RTD uses:

- A) Semiconductor
 - B) Pure metal
 - C) Thermocouple
 - D) Thermistor
-

Q.27 [2 Marks] [MODERATE] [Measurement Systems — Instrument Types]

Strain gauge factor GF =:

- A) $\Delta R/R \div \Delta L/L$
 - B) $\Delta R/R \div \Delta L/L$
 - C) $\Delta R/R \times \Delta L/L$
 - D) $\Delta R/R \times \Delta L$
-

Q.28 [2 Marks] [MODERATE] [Control Systems — Root Locus]

CRO displays:

- A) Current vs Time

B) Voltage vs Time

C) Power vs Time

D) Frequency vs Time

Q.29 [2 Marks] [MODERATE] [Digital Electronics — Combinational Circuits]

LVDT measures:

- A) Displacement
- B) Velocity
- C) Acceleration
- D) Force

Q.30 [2 Marks] [MODERATE] [Analog Electronics — Amplifiers]

RTD uses:

- A) Semiconductor
- B) Pure metal
- C) Thermocouple
- D) Thermistor

Q.31 [2 Marks] [MODERATE] [Sensors & Transducers — Optical Sensors]

Strain gauge factor GF =:

- A) $\Delta L/L$
- B) $\Delta R/R$
- C) $\Delta R/R \div \Delta L/L$
- D) $\Delta R/L$

Q.32 [2 Marks] [MODERATE] [Measurement Systems — Errors]

CRO displays:

- A) Current vs Time
- B) Voltage vs Time
- C) Power vs Time
- D) Frequency vs Time

Q.33 [2 Marks] [HARD] [Control Systems — Block Diagrams]

LVDT measures:

- A) Displacement
- B) Velocity
- C) Acceleration
- D) Force

Q.34 [2 Marks] [HARD] [Digital Electronics — ADC/DAC]

RTD uses:

- A) Semiconductor
- B) Pure metal
- C) Thermocouple
- D) Thermistor

Q.35 [2 Marks] [HARD] [Analog Electronics — Filters]

Strain gauge factor GF =:

- A) $\Delta L/L$

B) 9DÂôÂ ò "R/R

C) 9E"ö" r "L/L

D) 9E"Ó"L

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Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Sensors & Transducers — Thermoelectric]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.37 [2 Marks] [HARD] [Measurement Systems — Errors]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.38 [2 Marks] [HARD] [Control Systems — PID Control]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.39 [2 Marks] [HARD] [Digital Electronics — Sequential Circuits]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Q.40 [2 Marks] [HARD] [Analog Electronics — Filters]

Which statements are correct?

- A) Statement A
 - B) Statement B
 - C) Statement C
 - D) Statement D
 - E) Statement E
-

Section C — Numerical Answer Type (NAT)**Q.51 [2 Marks] [HARD] [Sensors & Transducers — Optical Sensors]**

Wheatstone bridge: $R_1=R_2=R_3=R_4=100\Omega$ à Output voltage (V)?

Answer: _____

Q.52 [2 Marks] [HARD] [Measurement Systems — Instrument Types]

RTD: Pt100 at 0°C has resistance = ?

Answer: _____

Q.53 [2 Marks] [HARD] [Control Systems — Root Locus]

Wheatstone bridge: $R_1=R_2=R_3=R_4=100\Omega$ à Output voltage (V)?

Answer: _____

Q.54 [2 Marks] [HARD] [Digital Electronics — Microprocessors]

RTD: Pt100 at 0°C has resistance = ?

Answer: _____

Q.55 [2 Marks] [HARD] [Analog Electronics — Amplifiers]

Wheatstone bridge: $R_1=R_2=R_3=R_4=100\Omega$ à Output voltage (V)?

Answer: _____

Q.56 [2 Marks] [HARD] [Sensors & Transducers — Thermoelectric]

RTD: Pt100 at 0°C has resistance = ?

Answer: _____

Q.57 [2 Marks] [HARD] [Measurement Systems — Errors]

Wheatstone bridge: $R_1=R_2=R_3=R_4=100\Omega$ à Output voltage (V)?

Answer: _____

Q.58 [2 Marks] [HARD] [Control Systems — Root Locus]

RTD: Pt100 at 0°C has resistance = ?

Answer: _____

Q.59 [2 Marks] [HARD] [Digital Electronics — Microprocessors]

Wheatstone bridge: $R_1=R_2=R_3=R_4=100\Omega$ Output voltage (V)?

Answer: _____

Q.60 [2 Marks] [HARD] [Analog Electronics — Oscillators]

RTD: Pt100 at 0°C has resistance = ?

Answer: _____

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Answer Key

Verify your answers after completing the full paper.

Q.1:	A	Q.2:	B	Q.3:	A
Q.4:	B	Q.5:	A	Q.6:	B
Q.7:	A	Q.8:	B	Q.9:	A
Q.10:	B	Q.11:	A	Q.12:	B
Q.13:	A	Q.14:	B	Q.15:	A
Q.16:	B	Q.17:	A	Q.18:	B
Q.19:	A	Q.20:	B	Q.21:	A
Q.22:	B	Q.23:	A	Q.24:	B
Q.25:	A	Q.26:	B	Q.27:	A
Q.28:	B	Q.29:	A	Q.30:	B
Q.31:	A	Q.32:	B	Q.33:	A
Q.34:	B	Q.35:	A	Q.36:	AC
Q.37:	AC	Q.38:	AC	Q.39:	AC
Q.40:	AC	Q.41:	AC	Q.42:	AC
Q.43:	AC	Q.44:	AC	Q.45:	AC
Q.46:	AC	Q.47:	AC	Q.48:	AC
Q.49:	AC	Q.50:	AC	Q.51:	0
Q.52:	100	Q.53:	0	Q.54:	100
Q.55:	0	Q.56:	100	Q.57:	0
Q.58:	100	Q.59:	0	Q.60:	100

Detailed Solutions

Question 1 [Sensors & Transducers — Resistive/Capacitive]

LVDT measures:

Answer: (A)

LVDT = Linear Variable Differential Transformer. Measures linear displacement.

Question 2 [Measurement Systems — Bridge Circuits]

RTD uses:

Answer: (B)

RTD uses pure metal (Pt, Ni) with positive temperature coefficient.

Question 3 [Control Systems — Compensators]Strain gauge factor $GF =$:**Answer:** (A) $GF = (9E''\delta''' \div f''L/L)$. Gauges strain in conductors.

Question 4 [Digital Electronics — ADC/DAC]

CRO displays:

Answer: (B)

Cathode Ray Oscilloscope displays voltage vs time.

Question 5 [Analog Electronics — Oscillators]

LVDT measures:

Answer: (A)

LVDT = Linear Variable Differential Transformer. Measures linear displacement.

Question 6 [Sensors & Transducers — Resistive/Capacitive]

RTD uses:

Answer: (B)

RTD uses pure metal (Pt, Ni) with positive temperature coefficient.

Question 7 [Measurement Systems — CRO]Strain gauge factor $GF =$:**Answer:** (A) $GF = (9E''\delta''' \div f''L/L)$. Gauges strain in conductors.

Question 8 [Control Systems — Root Locus]

CRO displays:

Answer: (B)

Cathode Ray Oscilloscope displays voltage vs time.

Question 9 [Digital Electronics — Logic Gates]

LVDT measures:

Answer:

LVDT = Linear Variable Differential Transformer. Measures linear displacement.

Question 10 [Analog Electronics — Op-Amps]

RTD uses:

Answer: (B)

RTD uses pure metal (Pt, Ni) with positive temperature coefficient.

Question 11 [Sensors & Transducers — Thermoelectric]

Strain gauge factor $GF =$:

Answer: (A)

$GF = (9E''\delta''' \div f''L/L)$. Gauges strain in conductors.

Question 12 [Measurement Systems — CRO]

CRO displays:

Answer: (B)

Cathode Ray Oscilloscope displays voltage vs time.

Question 13 [Control Systems — PID Control]

LVDT measures:

Answer: (A)

LVDT = Linear Variable Differential Transformer. Measures linear displacement.

Question 14 [Digital Electronics — Sequential Circuits]

RTD uses:

Answer: (B)

RTD uses pure metal (Pt, Ni) with positive temperature coefficient.

Question 15 [Analog Electronics — Op-Amps]

Strain gauge factor $GF =$:

Answer: (A)

$GF = (9E''\delta''' \div f''L/L)$. Gauges strain in conductors.

Question 16 [Sensors & Transducers — Optical Sensors]

CRO displays:

Answer: (B)

Cathode Ray Oscilloscope displays voltage vs time.

Question 17 [Measurement Systems — Errors]

LVDT measures:

Answer: (A)

LVDT = Linear Variable Differential Transformer. Measures linear displacement.

Question 18 [Control Systems — PID Control]

RTD uses:

Answer:

(B)

RTD uses pure metal (Pt, Ni) with positive temperature coefficient.

Question 19 [Digital Electronics — Sequential Circuits]

Strain gauge factor $GF =$:

Answer: (A)

$GF = (9E^{-6} \Delta L / L)$. Gauges strain in conductors.

Question 20 [Analog Electronics — Diodes]

CRO displays:

Answer: (B)

Cathode Ray Oscilloscope displays voltage vs time.

Question 21 [Sensors & Transducers — Thermoelectric]

LVDT measures:

Answer: (A)

LVDT = Linear Variable Differential Transformer. Measures linear displacement.

Question 22 [Measurement Systems — Signal Conditioning]

RTD uses:

Answer: (B)

RTD uses pure metal (Pt, Ni) with positive temperature coefficient.

Question 23 [Control Systems — PID Control]

Strain gauge factor $GF =$:

Answer: (A)

$GF = (9E^{-6} \Delta L / L)$. Gauges strain in conductors.

Question 24 [Digital Electronics — Logic Gates]

CRO displays:

Answer: (B)

Cathode Ray Oscilloscope displays voltage vs time.

Question 25 [Analog Electronics — Amplifiers]

LVDT measures:

Answer: (A)

LVDT = Linear Variable Differential Transformer. Measures linear displacement.

Question 26 [Sensors & Transducers — Thermoelectric]

RTD uses:

Answer: (B)

RTD uses pure metal (Pt, Ni) with positive temperature coefficient.

Question 27 [Measurement Systems — Instrument Types]

Strain gauge factor $GF =$:

Answer:

$GF = (9E''\delta''' \div f''L/L)$. Gauges strain in conductors.

Question 28 [Control Systems — Root Locus]

CRO displays:

Answer: (B)

Cathode Ray Oscilloscope displays voltage vs time.

Question 29 [Digital Electronics — Combinational Circuits]

LVDT measures:

Answer: (A)

LVDT = Linear Variable Differential Transformer. Measures linear displacement.

Question 30 [Analog Electronics — Amplifiers]

RTD uses:

Answer: (B)

RTD uses pure metal (Pt, Ni) with positive temperature coefficient.

Question 31 [Sensors & Transducers — Optical Sensors]

Strain gauge factor $GF =$:

Answer: (A)

$GF = (9E''\delta''' \div f''L/L)$. Gauges strain in conductors.

Question 32 [Measurement Systems — Errors]

CRO displays:

Answer: (B)

Cathode Ray Oscilloscope displays voltage vs time.

Question 33 [Control Systems — Block Diagrams]

LVDT measures:

Answer: (A)

LVDT = Linear Variable Differential Transformer. Measures linear displacement.

Question 34 [Digital Electronics — ADC/DAC]

RTD uses:

Answer: (B)

RTD uses pure metal (Pt, Ni) with positive temperature coefficient.

Question 35 [Analog Electronics — Filters]

Strain gauge factor $GF =$:

Answer: (A)

$GF = (9E''\delta''' \div f''L/L)$. Gauges strain in conductors.

Question 36 [Sensors & Transducers — Thermoelectric]

Which statements are correct?

Answer:

Based on theoretical analysis of the topic.

Question 37 [Measurement Systems — Errors]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 38 [Control Systems — PID Control]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 39 [Digital Electronics — Sequential Circuits]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 40 [Analog Electronics — Filters]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 41 [Sensors & Transducers — Digital Transducers]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 42 [Measurement Systems — Signal Conditioning]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 43 [Control Systems — Block Diagrams]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 44 [Digital Electronics — Logic Gates]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 45 [Analog Electronics — Filters]

Which statements are correct?

Answer:

Based on theoretical analysis of the topic.

Question 46 [Sensors & Transducers — Optical Sensors]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 47 [Measurement Systems — Instrument Types]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 48 [Control Systems — Root Locus]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 49 [Digital Electronics — Sequential Circuits]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 50 [Analog Electronics — Oscillators]

Which statements are correct?

Answer: (A, C)

Based on theoretical analysis of the topic.

Question 51 [Sensors & Transducers — Optical Sensors]

Wheatstone bridge: $R_1=R_2=R_3=R_4=100\Omega$ → Output voltage (V)?

Answer: (0)

Balanced bridge: $R_1/R_2 = R_4/R_3 = 1$. Output = 0.

Question 52 [Measurement Systems — Instrument Types]

RTD: Pt100 at 0°C has resistance = ?

Answer: (100)

Pt100 = 100 Ω B 0. Coefficient α_H 0.385 $\pm$ 0.005.

Question 53 [Control Systems — Root Locus]

Wheatstone bridge: $R_1=R_2=R_3=R_4=100\Omega$ → Output voltage (V)?

Answer: (0)

Balanced bridge: $R_1/R_2 = R_4/R_3 = 1$. Output = 0.

Question 54 [Digital Electronics — Microprocessors]

RTD: Pt100 at 0°C has resistance = ?

Answer:

Pt100 = 100:° B 0. Coefficient "H 0.385 :°û 0.

Question 55 [Analog Electronics — Amplifiers]

Wheatstone bridge: $R_1=R_2=R_3=R_4=100\text{:}^\circ$ à Output voltage (V)?

Answer: (0)

Balanced bridge: $R_1/R_2 = R_4/R_3 = 1$. Output = 0.

Question 56 [Sensors & Transducers — Thermoelectric]

RTD: Pt100 at 0°C has resistance = ?

Answer: (100)

Pt100 = 100:° B 0. Coefficient "H 0.385 :°û 0.

Question 57 [Measurement Systems — Errors]

Wheatstone bridge: $R_1=R_2=R_3=R_4=100\text{:}^\circ$ à Output voltage (V)?

Answer: (0)

Balanced bridge: $R_1/R_2 = R_4/R_3 = 1$. Output = 0.

Question 58 [Control Systems — Root Locus]

RTD: Pt100 at 0°C has resistance = ?

Answer: (100)

Pt100 = 100:° B 0. Coefficient "H 0.385 :°û 0.

Question 59 [Digital Electronics — Microprocessors]

Wheatstone bridge: $R_1=R_2=R_3=R_4=100\text{:}^\circ$ à Output voltage (V)?

Answer: (0)

Balanced bridge: $R_1/R_2 = R_4/R_3 = 1$. Output = 0.

Question 60 [Analog Electronics — Oscillators]

RTD: Pt100 at 0°C has resistance = ?

Answer: (100)

Pt100 = 100:° B 0. Coefficient "H 0.385 :°û 0.
